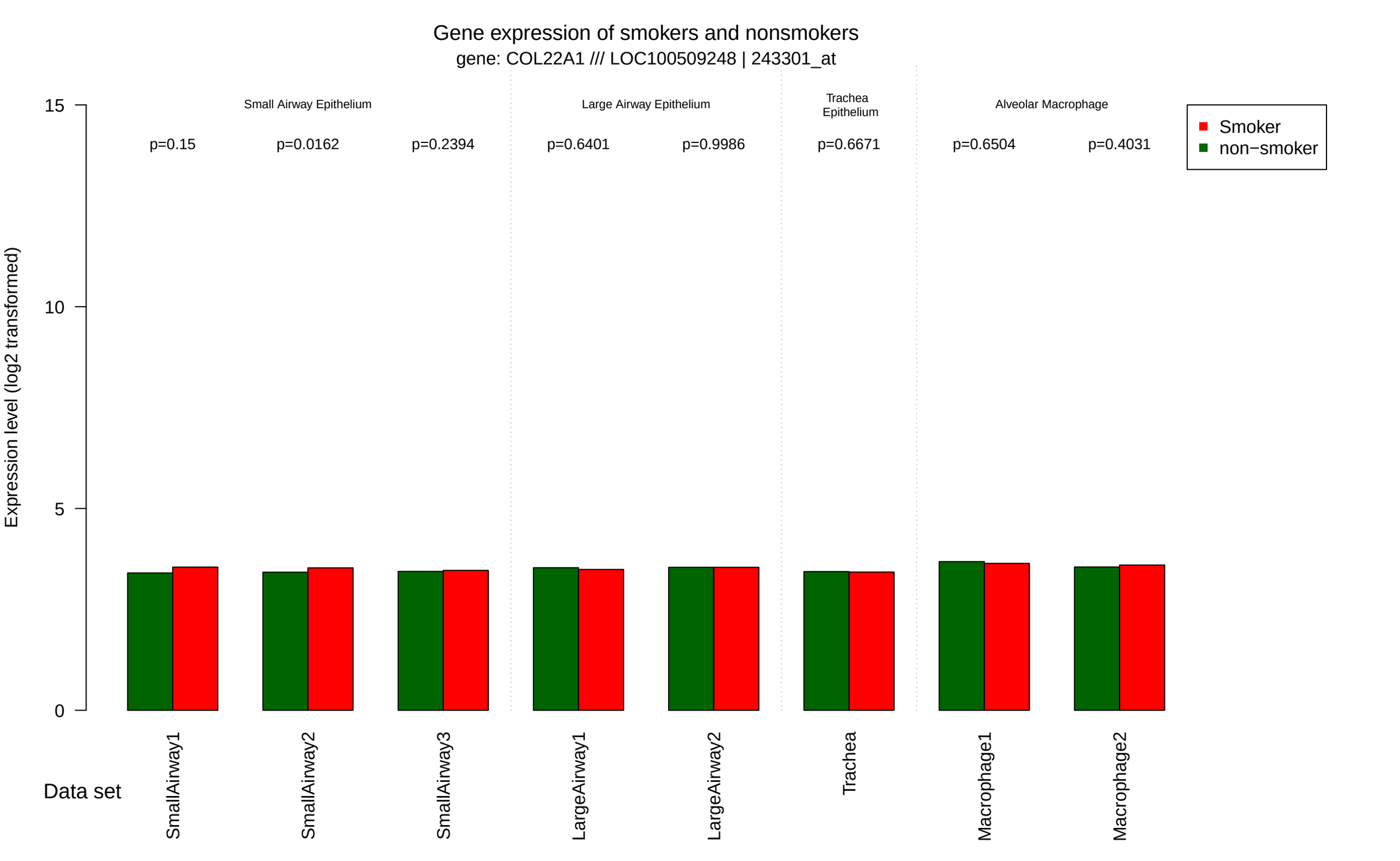
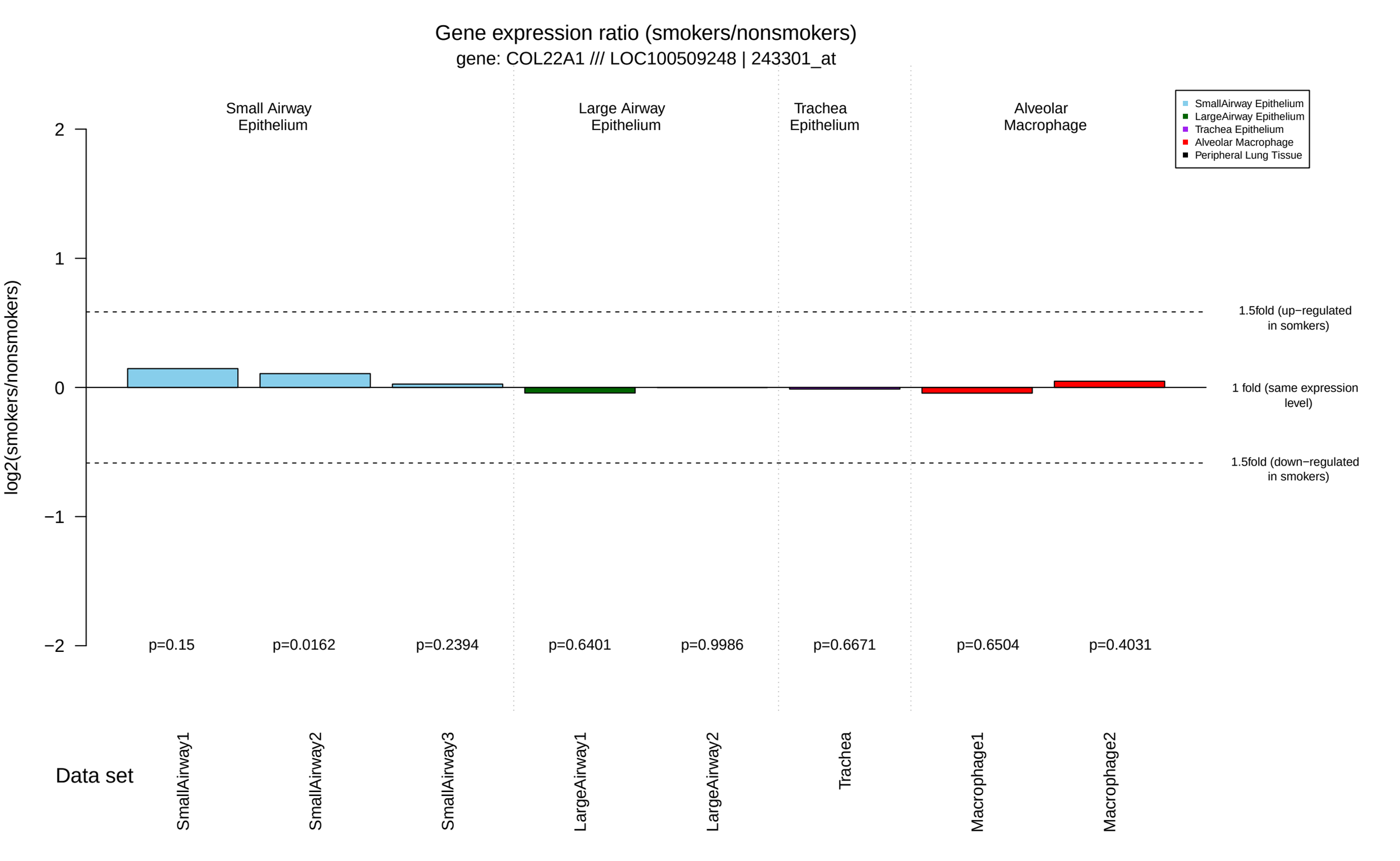
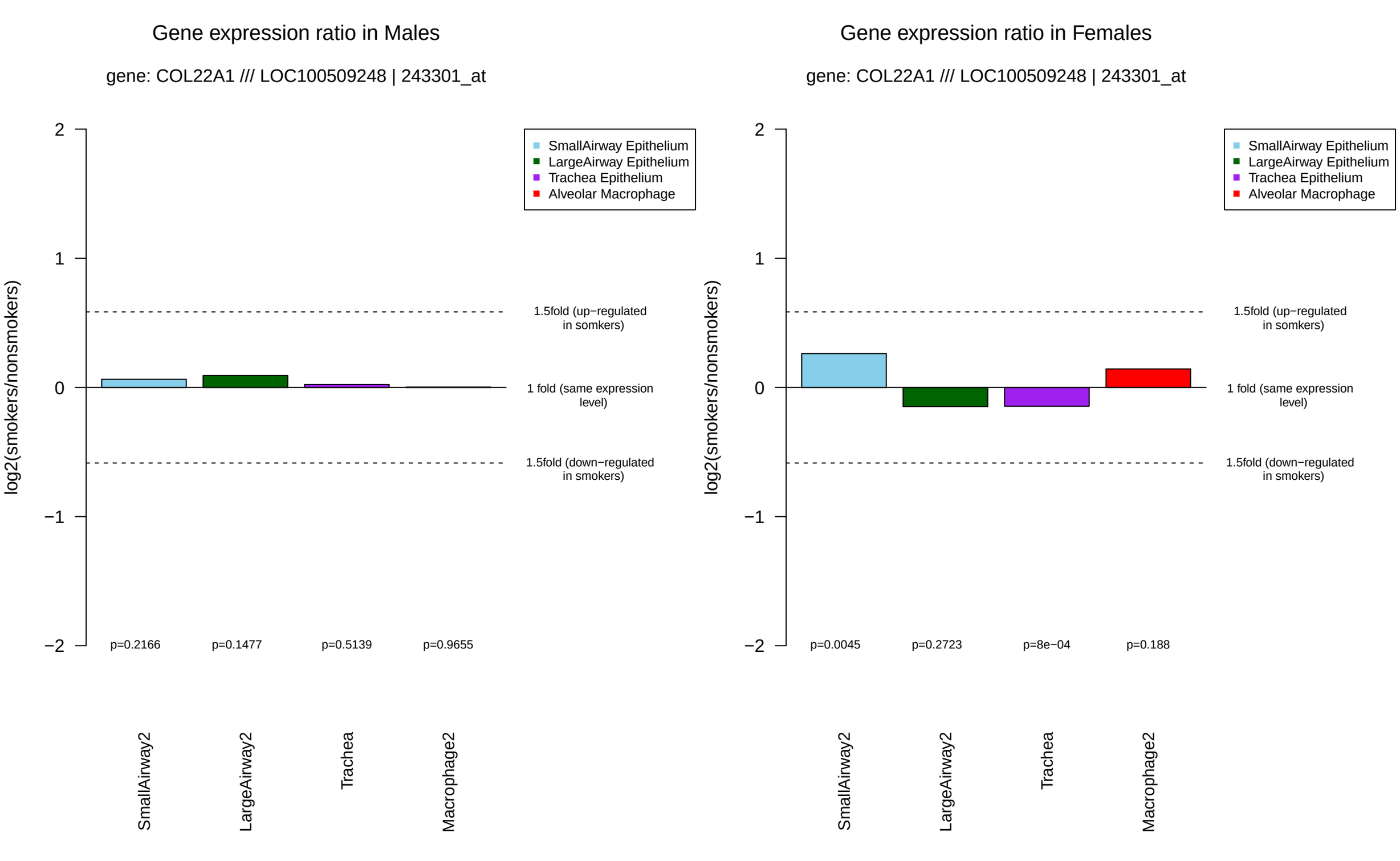
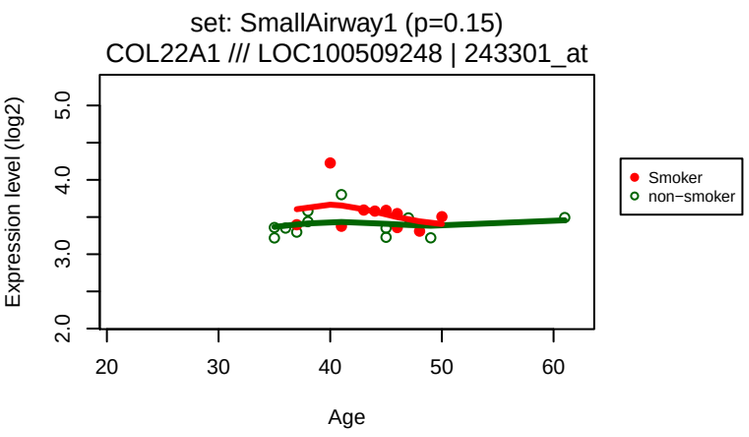
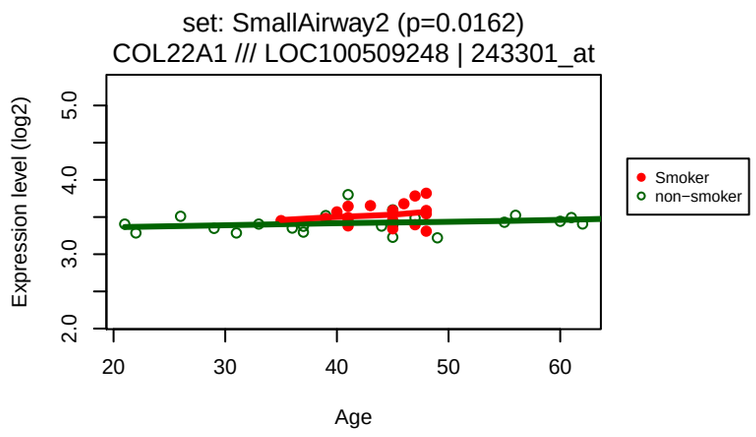
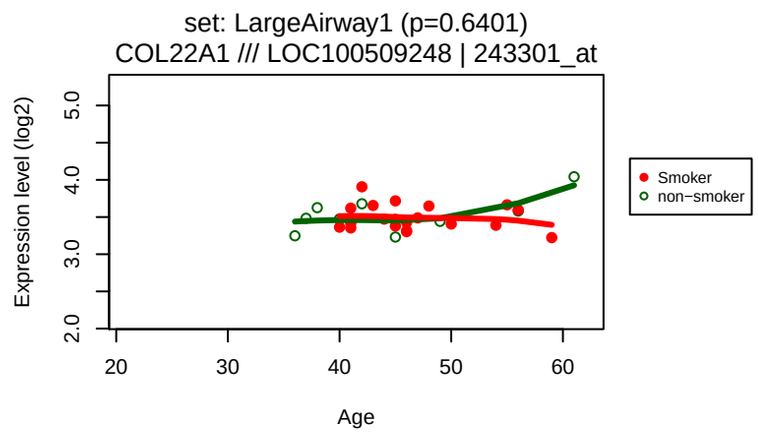

Gene expression of smokers and nonsmokers
gene: COL22A1 /// LOC100509248 | 243301_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.15
p=0.0162
p=0.2394
p=0.6401
p=0.9986
p=0.6671
p=0.6504
p=0.4031
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

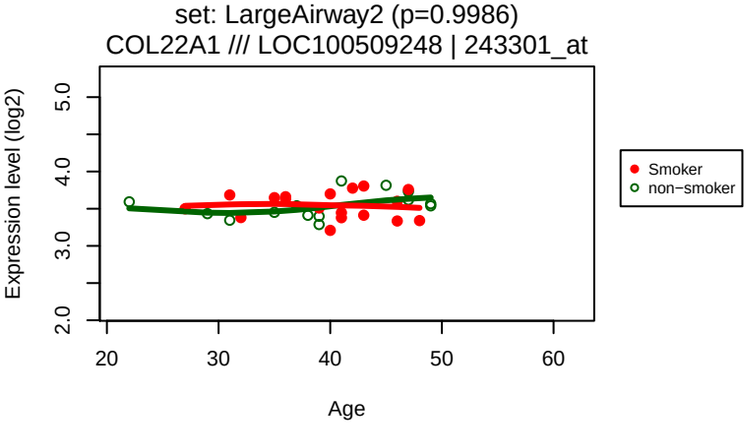

Gene expression ratio (smokers/nonsmokers)
gene: COL22A1 /// LOC100509248 | 243301_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.15
p=0.0162
p=0.2394
p=0.6401
p=0.9986
p=0.6671
p=0.6504
p=0.4031
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

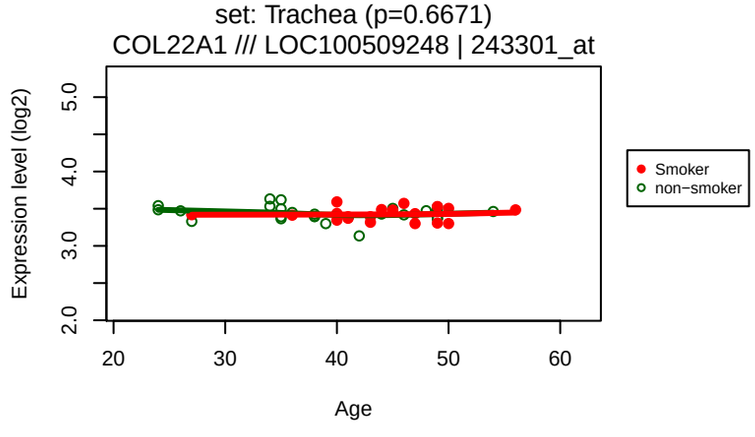

Gene expression ratio in Males
gene: COL22A1 /// LOC100509248 | 243301_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2166
p=0.1477
p=0.5139
p=0.9655
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL22A1 /// LOC100509248 | 243301_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0045
p=0.2723
p=8e−04
p=0.188
SmallAirway2
LargeAirway2
Trachea
Macrophage2

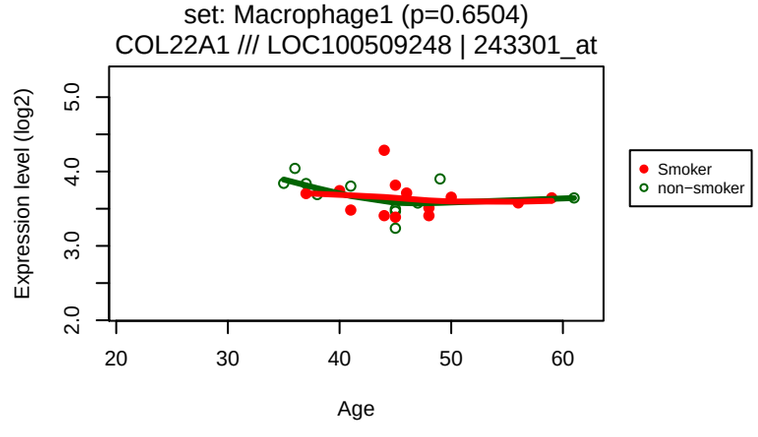

set: SmallAirway1 (p=0.15)
COL22A1 /// LOC100509248 | 243301_at
Expression level (log2)
2.0
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker

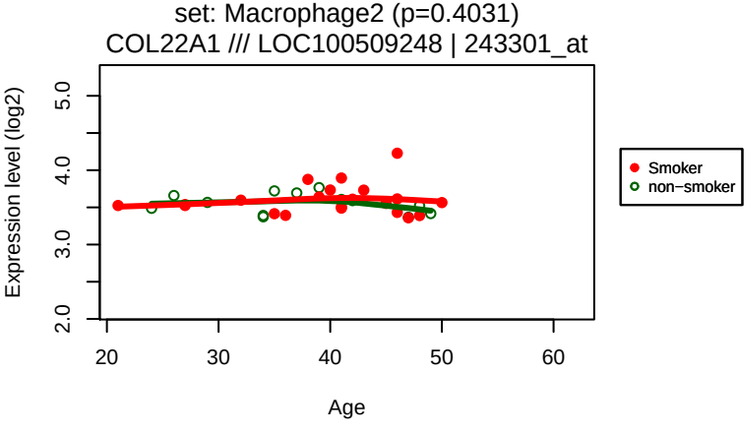

set: SmallAirway2 (p=0.0162)
COL22A1 /// LOC100509248 | 243301_at
Expression level (log2)
2.0
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker

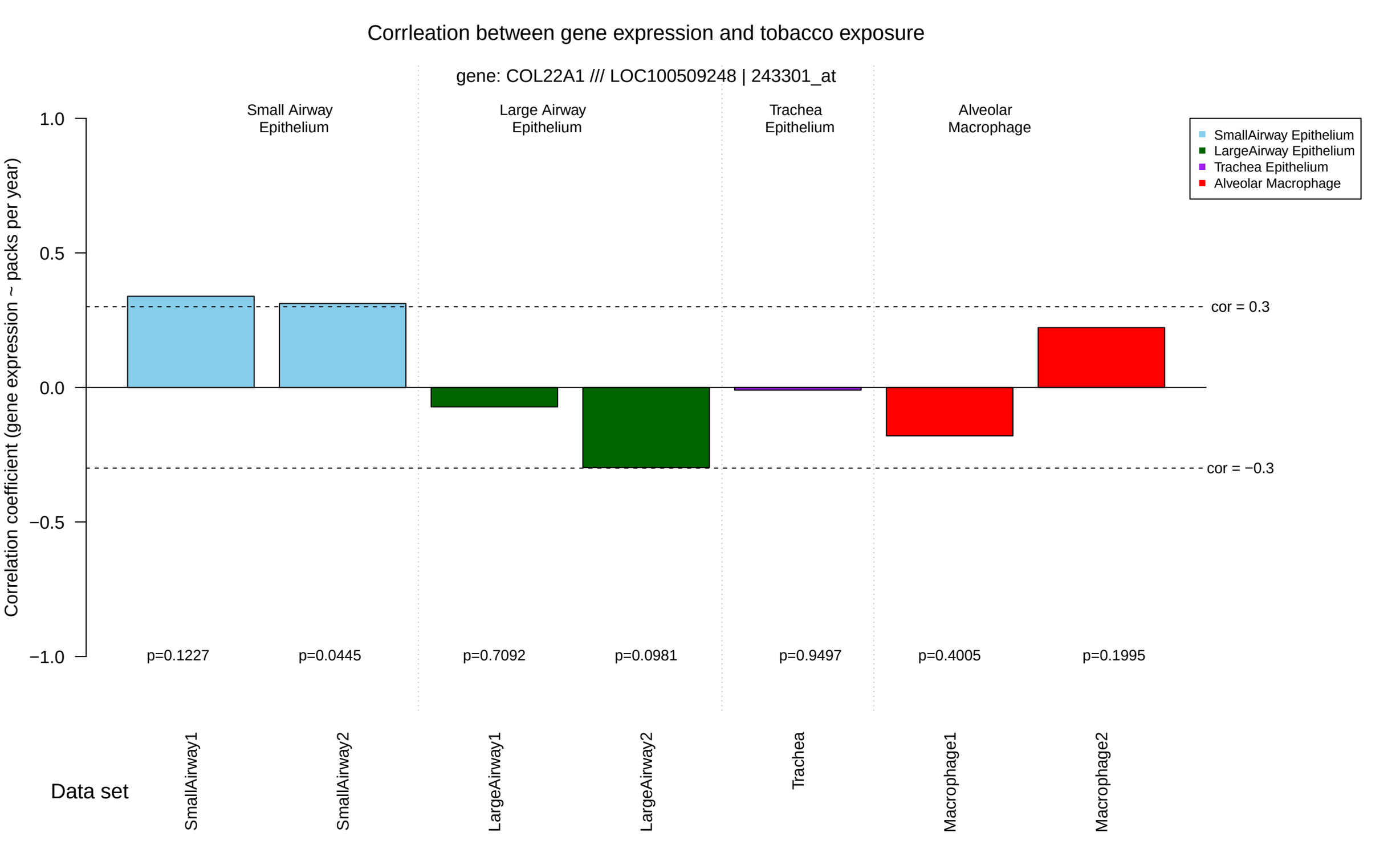

set: LargeAirway1 (p=0.6401)
COL22A1 /// LOC100509248 | 243301_at
Expression level (log2)
2.0
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9986)
COL22A1 /// LOC100509248 | 243301_at
Expression level (log2)
2.0
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.6671)
COL22A1 /// LOC100509248 | 243301_at
Expression level (log2)
2.0
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6504)
COL22A1 /// LOC100509248 | 243301_at
Expression level (log2)
2.0
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4031)
COL22A1 /// LOC100509248 | 243301_at
Expression level (log2)
2.0
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL22A1 /// LOC100509248 | 243301_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1227
p=0.0445
p=0.7092
p=0.0981
p=0.9497
p=0.4005
p=0.1995
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2